

Subhan Care HMS

Sprint Report — MERN Backend Integration

Prepared 15 July 2026

Summary

This sprint replaced the hospital management system's mock-data frontend with a real, persistent MERN backend (MongoDB, Express, React, Node.js) across five core modules — Appointments, Prescriptions, Medical History, Inventory, and Billing. Three cross-module automation features, PDF export, advanced search/filtering, and a new Reports & Analytics dashboard were also delivered.

New backend service	Node.js / Express / MongoDB (Mongoose), JWT authentication
Modules wired to live data	Appointments, Prescriptions, Medical History, Inventory, Billing
New collections/models	8 (User, Patient, Doctor, Appointment, Prescription, Medical History, Inventory Item, Invoice)
REST API endpoints	30+, matching the frontend's existing service-layer contract
Cross-module features	3 (auto inventory deduction, appointment -> prescription, appointment -> invoice)
New capabilities	PDF export, multi-field date-range search, Reports & Analytics dashboard

Work Completed

Backend Foundation

- Built the Express application: config, models, controllers, routes, and middleware layers.
- Implemented JWT-based authentication (login, forgot password / OTP, reset password, session rehydration).
- Server-side role-permission engine mirroring the app's 5 roles (Admin, Doctor, Receptionist, Pharmacist, Billing) as the single source of truth, replacing the old client-only mock.
- Built a seed script that populates the database from the app's existing demo data set, including all 5 demo login accounts.

Appointments, Prescriptions & Medical History

- Appointments, Prescriptions, and Medical History pages now read and write through the real API instead of local mock arrays.
- Live patient and doctor pickers replace hardcoded dropdown lists.
- Medical History entries remain strictly append-only and immutable, enforced at both the API routing layer and the database schema layer.
- Doctors are scoped server-side to their own prescriptions and appointments — enforced by the backend, not just hidden in the UI.

Inventory, Billing & Cross-Module Automation

- Inventory and Billing pages wired to the real API with full create/update/delete flows.

- Automated stock deduction: dispensing a prescription now matches its medications against inventory and decrements stock automatically, with a safe fallback warning if no confident match is found.
- "Write Prescription" quick action appears on a doctor's completed appointments, carrying the patient through to the Prescriptions page.
- Billing staff can generate a draft invoice directly from a completed appointment via a new "From Appointment" picker.

PDF Export, Search/Filter & Analytics

- One-click PDF export/download for both invoices and prescriptions.
- A reusable date-range and multi-field search/filter capability was added across Appointments, Prescriptions, Inventory, and Billing.
- New Reports & Analytics page: revenue over time, inventory usage by category, prescription trends with top medications, and appointment load — all backed by live server-side aggregation and rendered with interactive charts.

Current Status & Next Steps

All backend and frontend code has been written and verified via a full production build and lint pass (zero errors, zero new warnings). The screenshots in this report were captured against a local demo dataset to confirm every screen renders and functions correctly end to end. Connecting to a persistent MongoDB Atlas database for live/production use is the remaining step before this can be deployed.

Outstanding before this can be deployed live:

- Finalize MongoDB Atlas connection (database credentials / network access).
- Run the seed script against the live database and complete a final end-to-end smoke test of each role's workflow.
- Review and commit the outstanding changes.

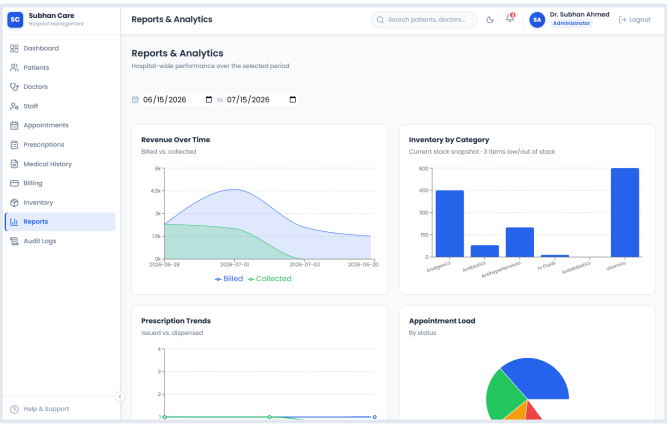
Scope Note

Two small, deliberate permission adjustments were made beyond the original module list: doctors and billing staff were given read-only access to the Appointments module (previously fully blocked), since the approved cross-module features require them to see their own relevant appointments. This was confirmed before implementation.

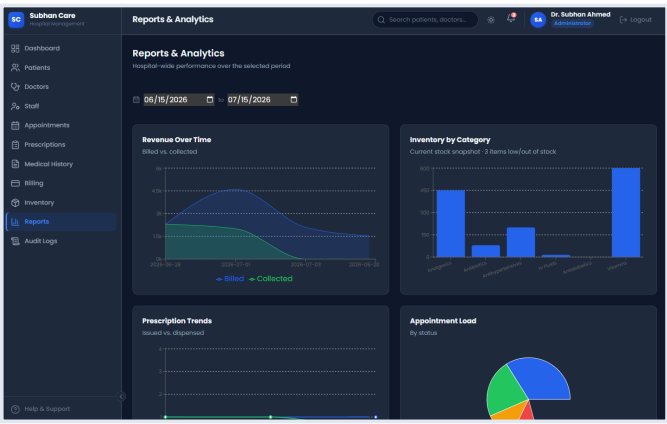
Reports & Analytics

New page — revenue, inventory usage, prescription trends, and appointment load, all as interactive charts.

Light mode



Dark mode



Billing & Payments

Live invoices with the new "From Appointment" quick-create action and a date-range filter.

Light mode

Subhan Care

Registration Management

Dashboard

Patients

Doctors

Staff

Appointments

Prescriptions

Medical History

Billing

Inventory

Reports

Audit Logs

Dr. Subhan Ahmed

Administrator

Logout

Billing & Payments

Search patients, doctors...

3 overdue invoices require immediate attention.

Total Collected

Rs. 4,300

Outstanding

Rs. 6,200

Overdue

3 Invoices

Invoices

4 total 1 paid

From Appointment

Create Invoice

Search by patient, invoice ID, or service...

All Statuses

mm/dd/yyyy

mm/dd/yyyy

INVOICE	PATIENT	DATE	DUE DATE	TOTAL	PAID	OUTSTANDING	STATUS	ACTIONS
INV-001	Ahmed Khan	2026-06-28	2026-07-05	Rs. 2,300	Rs. 2,300	Rs. 0	Paid	
INV-002	Sara Malik	2026-07-01	2026-07-08	Rs. 4,600	Rs. 2,000	Rs. 2,600	Overdue	
INV-003	Hassan Raza	2026-07-03	2026-07-10	Rs. 2,100	Rs. 0	Rs. 2,100	Overdue	
INV-004	Maryam Iqbal	2026-06-30	2026-06-27	Rs. 1,500	Rs. 0	Rs. 1,500	Overdue	

Help & Support

Dark mode

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INV-003	Hassan Raza	2026-07-03	2026-07-10	Rs. 2,100	Rs. 0	Rs. 2,100	Overdue	
INV-004	Maryam Iqbal	2026-06-30	2026-06-27	Rs. 1,500	Rs. 0	Rs. 1,500	Overdue	

Help & Support

Appointment Scheduling

Live doctor/patient data with multi-field search and date-range filtering.

Light mode

SC Subhan Care

Appointment Scheduling

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Help & Support

Appointment Scheduling

0 today · 2 pending confirmation

Search by patient or doctor...

All Doctors

All Statuses

mm/dd/yyyy

mm/dd/yyyy

+ Book Appointment

PATIENT	DOCTOR	DATE & TIME	TYPE	STATUS	ACTIONS
AK Ahmed Khan	Dr. Fatima Noor	2028-07-03 09:00	Consultation	Confirmed	
SM Sara Malik	Dr. Usman Ali	2028-07-03 09:00	Follow-up	Confirmed	
HR Hassan Raza	Dr. Ayesha Tariq	2028-07-03 09:00	Check-up	Pending	
MY Maryam Iqbal	Dr. Sara Riaz	2028-07-03 09:00	Consultation	Completed	
BC Bilal Chaudhry	Dr. Fatima Noor	2028-07-04 09:00	Emergency	Pending	
NH Nida Hussain	Dr. Usman Ali	2028-07-04 09:00	Follow-up	Cancelled	
TS Tariq Butt	Dr. Bilal Mahmood	2028-07-05 09:00	Consultation	Confirmed	

Dark mode

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Appointment Scheduling

Search patients, doctors...

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Inventory Management

Real-time stock levels, automatically decremented when prescriptions are dispensed.

Light mode

SC

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3 items are low on stock or out of stock — please reorder.

Inventory Management

6 items - 3 need attention

Search by name or supplier...

All Categories

All Statuses

mm/dd/yyyy

to

mm/dd/yyyy

ID	NAME	CATEGORY	QUANTITY	REORDER AT	UNIT RATE	EXPIRY	STATUS	ACTIONS
MS-001	Paracetamol 500mg	Analgesics	450 Tablets	100	Rs. 2.5	2027-12-31	Confirmed	
MS-002	Amoxicillin 500mg	Antibiotics	80 Capsules	100	Rs. 15	2026-09-30	Pending	
MS-003	Atorvastatin 10mg	Antihypertensives	200 Tablets	50	Rs. 8	2027-06-30	Confirmed	
MS-004	Normal Saline 0.9%	IV Fluids	15 Bottles	30	Rs. 80	2027-03-31	Pending	
MS-005	Insulin Glargine	Antidiabetics	0 Vials	10	Rs. 450	2026-09-30	Cancelled	
MS-006	Vitamin C 500mg	Vitamins	600 Tablets	100	Rs. 3	2028-01-31	Confirmed	

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MS-004	Normal Saline 0.9%	IV Fluids	15 Bottles	30	Rs. 80	2027-03-31	Pending	
MS-005	Insulin Glargine	Antidiabetics	0 Vials	10	Rs. 450	2026-09-30	Cancelled	
MS-006	Vitamin C 500mg	Vitamins	600 Tablets	100	Rs. 3	2028-01-31	Confirmed	

Help & Support

Prescription Management

Live prescriptions with one-click PDF download and doctor/date filters.

Light mode

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Prescription Management

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All prescriptions - 1 pending

Search by patient or diagnosis...

All Statuses

All Doctors

mm/dd/yyyy

mm/dd/yyyy

Rx ID	Patient	Doctor	Date	Diagnosis	Medications	Status	Actions
101	Ahmed Khan	Dr. Fatima Noor	2025-09-28	Hypertension, Hyperlipidemia	Aspirin 75mg, Atorvastatin 10mg	Dispensed	Download
102	Sara Malik	Dr. Usman Ali	2025-07-01	Post-operative pain management	Ibuprofen 400mg	Dispensed	Download
103	Hassan Raza	Dr. Fatima Noor	2025-07-03	Viral upper respiratory infection	Paracetamol 500mg, Chlorpheniramine 4mg	Pending	Download

Dark mode

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Prescription Management

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All prescriptions - 1 pending

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